Chapter 7
Bivalence, Future Contingents and the Open Future



Abstract In this chapter we critically discuss the objection that since truths require grounds, the Growing Block Theory must take bivalence to fail for future contingents, while it proves at odds with the best account of such a failure. We challenge the version of the grounding requirement driving this objection, devise a better formulation, and show that the theory can retain bivalence and accommodate an interesting form of indeterminism. After rehearsing the objection in Sect. 7.1, in Sect. 7.2 we review different ways to articulate the grounding requirement, conclude that it should suffice that, for any tensed truth, sometimes there be grounds for it, and show how this requirement can be met by contingent truths about the future. In Sect. 7.3 we explicate a conception of the asymmetry between the open future and the fixed past, consistent with bivalence and available to the Growing Block Theory but none of its rivals.

In this chapter, we will formulate another challenge to GBT and then likewise defuse it. Throughout it is being presupposed that there are *future contingents*, i.e. statements about the future whose truth-value is not predetermined by the present or past – a presupposition that we are happy to make. The challenge then proceeds from a particular formulation of the grounding requirement on truth that would seem to force proponents of GBT to reject the principle of bivalence and to hold that future contingents are neither true nor false. It is then argued that the only sensible account of this failure of bivalence for future contingents – supervaluationism – is at odds with central tenets of GBT. We call this *the semantic objection* to GBT. Subsequently, we critically discuss the particular formulation of the grounding requirement on tensed truth that drives the objection, and propose an alternative, in our view much more appropriate formulation that proves to be compatible with both GBT and presentism. Thus, contrary to common conception, it so turns out that as far as the grounding of truth is concerned, the commitment to past things does not put GBT in any better position to account for the truth of statements about the past. Finally, we show to what extent GBT's acceptance of bivalence still allows for conceptions of the open future that are stronger than traditional forms of indeterminism and unavailable to permanentists: the future may be said to be open because time could come to an end, with no ontological commitment to future things standing in the way. While presentists can likewise opt for this conception of openness, they can

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F. Correia, S. Rosenkranz, *Nothing To Come*, Synthese Library 395,
https://doi.org/10.1007/978-3-319-78704-6_7

only do so at the cost of jettisoning the asymmetry between the open future and the fixed past: for them, the past will be open in the corresponding sense in which time could just have begun, with no ontological commitment to past things standing in the way.

7.1 The Semantic Objection

Truths do not float free. They require grounds. This is easily seen, once we reflect upon the fact that saying something true is an achievement: whether our attempts to represent what the world is like are crowned with success depends on whether the world is the way we represent it as being. Accordingly, truths must be grounded in reality, because true representation is successful representation in this sense. The idea that truths must be grounded in order to qualify as truths is often put by saying that *all truths*, qua truths, *are grounded in what there is and how it is*. As we shall see in due course, this formulation must be taken with some care, as soon as we here understand the occurrences of 'are' and 'is' as being, without exception, in the present tense.

Consider a pair of future-tensed statements – respectively of the forms Fp and F$\neg p$, or the forms $F_n p$ and $F_n \neg p$, where n measures temporal distance and $F_n\varphi$ is understood in such a way as to entail Fφ – whose present truth-values are not already settled by how things located in the present or past of now are, or have been, in all their natural – non-future-directed, non-Cambridge-like – respects. For example, assume that both 'One day hence, some rain will fall' and 'One day hence, no rain will fall' belong to this category of statements. Statements of this category qualify as *future contingents* in that, even if one of them should presently be true, its present truth would anyway not be made inevitable, or be *predetermined*, by facts that are, strictly speaking, facts about what goes on in the present or what went on in the past.

That future contingents are not predetermined to be true in this sense does not *ipso facto* preclude that they are now both true and grounded. It is just that their truth, if they should be true, would not then be grounded in how things located in the present or past of now are or have been in all their natural, non-future-directed respects. Instantiations of Cambridge-like properties are unlikely candidates for being grounds for truth. Accordingly, if anything *now* grounds the present truth of 'One day hence, some rain will fall', it would either have to be something located in the present or past of now that has, or has had, some purely future-directed property – say, water-molecules, now hanging in a cloud, that presently have the historically contingent property of going to form raindrops one day hence – or something located entirely in the future of now like the event of tomorrow's rainfall. (Evidently, it is no good appealing to *causal* properties of things located in the present or past, apt to nomologically necessitate future facts and thereby to render the relevant statements already true now, since this would undermine the status of those statements as genuine future contingents.)

Suppose we follow Broad and assume an ontology of things in time exclusively composed of times and events. Then the first option is foreclosed, as no *event* located in the present or past of now is going to be a rainfall a day hence. It would clearly be cheating to say that the truth of 'One day hence, some rain will fall' is grounded in the fact that today's sunshine has the property of being such that, one day hence, some rain will fall: if anything, this is a Cambridge-like property of today's sunshine which can hardly be at the service of an account of *why* it is true to say that, one day hence, some rain will fall; and things are not getting any better, if we attribute such a property to the sum total of all events and times that are located in the present or past of now. Truth-grounding should not be said to be achieved by instantiations of such Cambridge-like properties of things located in the present or past.

Some philosophers have argued that *any* appeal to past- or future-directed properties – so-called Lucretian properties – amounts to cheating when it comes to showing that one's view complies with the grounding requirement on truth, never mind whether or not these properties are also Cambridge-like (Sider 2001: §2.3, Merricks 2007: 135). Thus, such philosophers likewise object to the idea that the truth of 'One day hence, some rain will fall' can be grounded in the fact that water-molecules that are now dispersed in a gaseous state presently have the property of going to form raindrops one day hence, or simply the fact that, one day hence, some rain will fall. The force of the objection against such tensed facts about the future is unclear. Thus, consider how an eternalist like Mellor (1998), who undertakes commitment to the existence of tenseless facts, will account for the truth of statements about the future such as 'One day hence, some rain will fall'. They will say that, if presently true, this statement is made presently true by the tenseless fact that it-is-raining-at-*t*, where *t* is some instant of tomorrow. This is not a case of cheating on anyone's count; and yet, the eternalist here invokes a tenseless fact that is, as it were, oriented towards the future. Why, then, are tensed theorists of time subject to the charge of cheating when they invoke the future-tensed fact that, one day hence, some rain will fall, to do the job? Being tensed, this fact does not always obtain, provided only that, on some days, it is sunny a day later. But why should this feature make one's appeal to the fact in question in any way objectionable in the current context?

All this is a matter of ongoing controversy, and we do not wish to pretend that the debate has been decided in anyone's favour (for a defence of Lucretianism, see Bigelow 1996). We here merely want to explore what follows from the assumption that Lucretianism is of no avail in this context. This would imply that the first option to account for the present truth of 'One day hence, some rain will fall' is indeed foreclosed, irrespective of whether one opts for an ontology of things in time exclusively composed of times and events.

However, according to GBT, neither does there presently *exist* anything that is located entirely in the future of now: tomorrow's rainfall does not now exist, and neither does tomorrow. So, the second option is likewise foreclosed. Consequently, there is, on this view, nothing at present such that facts about it could *presently* ground the truth of 'One day hence, some rain will fall'. This alone, surely, is not enough to establish that 'One day hence, no rain will fall' is presently true. Indeed,

insofar as tomorrow no rain falls, some other event will then occur incompatible with its raining tomorrow; and *ex hypothesi* no such event presently exists, and *a fortiori* no facts about it which could ground the truth of 'One day hence, no rain will fall'. Were present truths to require present grounding, proponents of GBT would accordingly be committed to conclude that neither 'One day hence, some rain will fall' nor 'One day hence, no rain will fall' is true. Quite generally, where Fφ and F¬φ are future contingents, proponents of GBT would then be forced to regard neither statement as true.

Even if the truth of the negation of a given statement just is the falsity of that statement, it does not yet follow from this that these future contingents are neither true nor false: 'One day hence, no rain will fall' is not the negation of 'One day hence, some rain will fall'. Generally, F¬φ is not the negation of Fφ, and so need not automatically be true whenever Fφ is false. What one would need in addition, in order to draw this further conclusion, is a principle like $\neg F\varphi \rightarrow F\neg\varphi$, or its metric counterpart, $\neg F_n\varphi \rightarrow F_n\neg\varphi$. These principles fail on so-called Peircean accounts according to which the future-tense operator is equivalent in meaning to 'Inevitably it will (in *n* units of time) be the case that'. For, if it is not inevitable that (one day hence) some rain will fall, this evidently does not entail that it is inevitable that (one day hence) no rain will fall. However, Peircean accounts are notoriously impoverished, as they leave us with no means at all to express the thought that something will be the case as a matter of mere historical contingency (cf. Prior 1967). In other words, we could not even formulate future contingents. In any case, this is not the understanding of the future-tense operator that our version of GBT, or Broad's version for that matter, presupposes.

Are we therefore bound to conclude that if all future contingents fail to be true, they likewise fail to be false? Not obviously so. The principles $\neg F\varphi \rightarrow F\neg\varphi$ and $\neg F_n\varphi \rightarrow F_n\neg\varphi$ are objectionable on other grounds. They in effect rule out that time has come to an end: if time has come to an end, then $\neg F\varphi$, $\neg F\neg\varphi$, $\neg F_n\varphi$ and $\neg F_n\neg\varphi$ should all hold. It is of no direct help to conditionalise these principles on the assumption that time goes on, for as long as it is deemed an open historical possibility that time has come to an end, this assumption will itself be a future contingent; and so if one initially thought that all future contingents are uniformly false, one will be able to accept the conditionalised principles already because one takes them to have a false antecedent. Unlike the unconditionalised principles, the conditionalised principles would accordingly seem to put no pressure on those who try to resist the conclusion that, just because they are untrue, future contingents are neither true nor false.

However, even if one cannot rule out, on purely logical grounds, that time has come to an end, one may still accept $\neg F\varphi \rightarrow F\neg\varphi$ and $\neg F_n\varphi \rightarrow F_n\neg\varphi$, or relevant instances of them, as historically necessary truths, because one thinks that something about the present and past makes it inevitable that time will go on indefinitely, in whatever way, or will at least go on for another day. In that case, one is indeed driven towards the conclusion that if neither 'One day hence, some rain will fall' nor 'One day hence, no rain will fall' is true, neither of them is false either.

That statements like ‘One day hence, some rain will fall’ and ‘One day hence, no rain will fall’ are neither true nor false is a conclusion that Broad himself explicitly draws. He contends that while present-tensed statements about the non-existent such as ‘Puck exists’ are uniformly false, statements about the future are without exception neither true nor false (Broad 1923: 70–73). His reason for such unequal treatment is as follows. According to Broad, those who assert the statement ‘Puck exists’ thereby assert ‘that some part of the existent has [the] set of characteristics’ ‘by which they [describe] Puck to themselves’, where this is rendered false by the negative fact that nothing exists that has those characteristics (Broad 1923: 71). By contrast, Broad maintains that the statement ‘Tomorrow will be wet’ can only be rendered true or false by the fact of tomorrow’s being wet or the fact of tomorrow’s being dry ‘when tomorrow comes’; and since tomorrow has not yet come, there is presently no fact that could render this statement either true or false (Broad 1923: 73).

If, in the statement ‘Tomorrow will be wet’, ‘tomorrow’ was meant to function as a singular term or a definite description taking wide scope over the future tense, then it is unobvious why Broad should not have said, instead, that the statement is false. After all, if anything, tomorrow is the day after today; and according to GBT, there is at present no such day. Thus, it seems a much more charitable interpretation to assume that, in Broad’s statement ‘Tomorrow will be wet’, the future tense takes wide scope over ‘tomorrow’, and that a more perspicuous, tense-logical rendition of that statement would be ‘It will be the case that the day after today is wet’, which, being of the form Fφ, is not relevantly different from our earlier ‘One day hence, some rain will fall’. Indeed, at a later stage of his discussion, Broad himself claims that what one *asserts* in making a statement such as ‘Tomorrow will be wet’ is ‘that the characteristic in question [here: being wet] will characterise some part of what will become [here: what will then be the next day]’, to assert which, Broad adds, is ‘compatible with the non-existence of the future’ (Broad 1923: 77).

Although we ultimately agree with Broad’s observation that the truth of such a future-tensed statement depends on how, in the future, matters will stand, there are at least two reasons for being dissatisfied with his narrative. First, if what is asserted by means of such a statement is that the characteristic in question *will* characterise something, why should the observation that nothing has as yet come to exist that has, or is going to have, that characteristic, in any way suggest that the statement is not yet either true or false?

Secondly, there is something singularly odd about Broad’s suggestion that a present statement about the future such as ‘It will be the case that the day after today is wet’ will be true or false only once tomorrow arrives.[1] What is certainly congenial to the tensed metaphysics that GBT provides is that one day hence, the present-tensed statement ‘The present day is wet’ will be rendered true or false by some fact, or event, that will not have rendered it true or false any earlier. If, as Broad

[1] This idea has been elaborated, in their respective ways, by Belnap et al. (2001) and MacFarlane (2003).

himself would seem to suggest, however qualifiedly so, always whatever is the case was going to be the case (Broad 1923: 78), the future truth of 'The present day is wet' one day hence should imply today's truth of 'It will be the case that the day after today is wet'. At least, there is no reason to deny this truth-value link, if there is no antecedent reason to claim that the statement presently lacks a truth-value. In any case, however, one would have expected that whatever the present truth of 'It will be the case that the day after today is wet' presently requires, *it is not the present existence of the kind of fact that a day hence, the truth of 'The present day is wet' will require*. Yet, if this is so, there is again no evident reason for thinking that just because that fact does not now exist, the future-tensed statement is not now true.

Broad would seem to draw a principled distinction between what the truth of a statement about the future would require – according to him, that there *is* a kind of fact that does not yet exist but will only ever exist in the future – and what it asserts – that there *will be* a fact of that kind. Yet, such a distinction should not impress as very natural and certainly requires argument. But even if it can be drawn, this does not render Broad's particular way of drawing it the least plausible: why should the truth of 'In the future, *p*' require the existence of the fact that *p*? If the fact that *p* presently existed, then plausibly, '*p*' would be true. Again, why should we be tempted to concede that the truth of 'In the future, *p*' is only ever to be had at the cost of '*p*' being already true? As will become clear in due course, similar sorts of considerations are suited to disarm the thought, sketched earlier, that insofar as GBT holds, future contingents fail the grounding requirement on truth, properly construed.

If one accepts, as Broad does, that future contingents present counterexamples to the principle of bivalence, one is accordingly faced with the question of what logic and semantics one ought to assume when reasoning about the open future. On this matter Broad remains frustratingly silent. The first option that comes to mind is to model truth-value gaps along the lines suggested by Kleene's or Łukasiewicz' three-valued logics. But these choices have well-known drawbacks. Even if bivalence fails because it is as yet open whether or not sometimes in the future, some rain falls, it is nonetheless desirable that $F\varphi \rightarrow F\varphi$ be valid. Yet, in Kleene's (strong) three-valued logic, if $F\varphi$ is neither true nor false, so is $F\varphi \rightarrow F\varphi$. On Łukasiewicz' logic, no comparable problem arises, because $F\varphi \rightarrow F\varphi$ will be true, even if $F\varphi$ is neither true nor false. However, on either choice of logic, $F\varphi \vee \neg F\varphi$ will be neither true nor false when $F\varphi$ is a future contingent, and so will be $\neg(F\varphi \,\&\, \neg F\varphi)$; and this is clearly undesirable: even if it may presently be unsettled whether or not some rain will fall, it should anyway be settled that either it is the case that some rain will fall, or it is not the case that some rain will fall, and that anyway not both is the case (cf. Prior 1953). It is for these reasons that *supervaluationism* seems to be a much better choice, in order to model the denial of bivalence for future contingents which, given the foregoing, proponents of GBT would seem to be committed to (Thomason 1970; cf. also MacFarlane 2003). For, the non-bivalent semantics that supervaluationism affords still underwrites all theorems of classical logic.

Supervaluationism assumes that time is forward-branching but not backward-branching. Thus, where we take the variables '*u*', '*v*' and '*w*' to range over times,

supervaluationism implies (i), which rules out backward branching, but not (ii), which, if it held, would rule out forward branching:

(i) $\forall u \forall v \forall w(v \prec u \,\&\, w \prec u \rightarrow (v \neq w \rightarrow v \prec w \vee w \prec v))$
(ii) $\forall u \forall v \forall w(u \prec v \,\&\, u \prec w \rightarrow (v \neq w \rightarrow v \prec w \vee w \prec v))$

According to supervaluationism, φ is true at a time t just in case φ is supertrue at t; and φ is false at time t just in case φ is superfalse at t. Supertruth, and superfalsity, at a given time t are defined as follows: φ is supertrue at time t just in case φ is true at t on all histories that include t; and φ is superfalse at time t just in case φ is false at t on all histories that include t. Here, a history is a maximal linear set of times, where a set s of times is linear iff $\forall u \forall v(u \in s \,\&\, v \in s \rightarrow (u \neq v \rightarrow u \prec v \vee v \prec u))$, and is a maximal such set iff there is no set of times s^* such that s^* is linear and s is strictly included in s^*. For all histories h and any $t \in h$, truth at $\langle t, h\rangle$ behaves classically; and Fφ is true at $\langle t, h\rangle$ iff there is a $t^* \in h$ such that $t \prec t^*$ and φ is true at $\langle t^*, h\rangle$. Accordingly, given how supertruth was just defined, for any φ, whether or not it is a future contingent, both $\varphi \vee \neg\varphi$ and $\neg(\varphi \,\&\, \neg\varphi)$ come out supertrue at any time, even if there are times at which neither φ nor ¬φ is supertrue. Similarly, $F\varphi \rightarrow F\varphi$ comes out supertrue at any time, even if there are times at which neither Fφ nor F¬φ is supertrue.

However, it should be clear that the supervaluationist treatment cannot be made to square with one of the consequences of GBT, *viz.* that there is no time later than now. For if now is the last time, there can be only one history, that history having a last moment, namely now. Accordingly, given the supervaluationist's semantic treatment of the future tense operator F and of the operators ¬ and &, *all* statements of type $\neg F\varphi \,\&\, \neg F\neg\varphi$ must be taken to be supertrue now, and hence to be true now. This means that given the supervaluationist view, GBT is committed to time having just come to an end. This is, of course, an unfortunate result.

Here then is, at last, the semantic objection to GBT: insofar as present truths need to be presently grounded in what there presently is and how it presently is, where the characterisation of how things presently are must make no appeal to any Cambridge-like or future-directed Lucretian properties, proponents of GBT are bound to treat future contingents as being neither true nor false, lest they be committed to assuming that time will not go on. However, the best way to accommodate the thought that future contingents are neither true nor false is to opt for supervaluationism: all other attempts to model truth-value gaps have unpalatable consequences, including rejection of the law of excluded middle. Yet, supervaluationism is ultimately at odds with central tenets of GBT.

7.2 The Grounding Requirement on Truth Revisited

The semantic objection to GBT heavily relies upon a particular formulation of the grounding requirement on truth, and tensed truths in particular. This formulation is not the only one we might give, though. Thus, there is a deflationary take on the grounding requirement that exploits Aristotle's insight that '[i]t is not because we think truly that you are pale, that you are pale, but because you are pale we who say this have the truth'.[2] The core idea of such a deflationary approach is that we can straightforwardly discharge the grounding requirement, properly understood, by simply using the statement, or articulating the proposition, whose truth is in question, in order to state what grounds its truth (Tallant and Ingram 2015). Suitably generalising from Aristotle's dictum, we arrive at a schematic principle according to which for any substitution instance for '*p*', '*p*' is true, if it is true, because *p* – where it is here being assumed as part of the logic of 'because' that claims having it as main connective behave asymmetrically: if 'ψ because φ' holds, then it is not the case that 'φ because ψ' holds (see Correia and Schnieder 2012: 8, 26).

According to this deflationary approach, that truths be grounded amounts to no more than that suitable instances of the Aristotelian schema should hold for the truths in question; and that these instances hold is taken to follow from an explication of the notion of truth itself. There is then no need to come up with any more informative specifications of the grounds for truths – let alone any specifications that relate to what presently exists, unless the truths in question are themselves existential claims. Thus, for the case of future contingents like 'One day hence, some rain will fall', this means that there is as yet no pressure at all to explain why they are true, if they are true, by appeal to the kinds of purely futurely located things whose existence GBT denies.

Although we have sympathies for this deflationary approach to truth-grounding, we recognize that it is likely to leave objectors with the feeling of having been shortchanged. After all, the question was precisely what in the reality of things and their properties and relations that alone GBT acknowledges to exist – i.e. in the block grown as far as the present – can possibly account for the truth of future contingents; and this question does not go away by helping oneself to those future contingents in explaining their truth *on condition that they are true at all*. Rather, this question immediately translates into one about whether that sort of explanation can ever be accurate, because there may just turn out to be nothing that would ever ground discharge of the assumption that such future contingents *are* true.

Let us therefore concede that more needs to be said about truth-grounding than the deflationary approach supplies. Let us also concede to the objector that truth-grounding is always in terms of what there is and how it is. These concessions are not yet an admission of defeat in one's attempt to reconcile GBT with the truth of future contingents. For as long as no case has been made that, in general, in order for a statement to be presently true, it must *presently* be grounded in what there is

[2] *Metaphysics*, 1051b6–8; the translation follows Ross et al. (1908).

and how it is, GBT has nothing to fear; and it is precisely such a case that we think just is not forthcoming.

To begin with, let us ask again: Why should the truth of a given statement require grounding by something that the statement itself does not even claim or assert to exist? According to GBT, a statement of the form 'A thousand years hence, a child will be born' – unlike a statement of the form 'Some child will be born a thousand years hence' or 'Something will a thousand years hence be a newly born child' – does not affirm the existence of anything. How do we explain this difference in content between these different types of statements, if we simultaneously maintain that the present truth of all of them alike requires the present existence of something that will a thousand years hence be a newly born child?

It is admittedly unclear what the force of these intuitive considerations is. Arguably, many truths are grounded in things that they are not, in any obvious sense, 'about' (*pace* Merricks 2007). Thus, on certain views, statements such as 'Putin is reckless' require the existence of tropes – e.g. Putin's particularized recklessness – in order to qualify as true, while it is certainly too far a stretch to suggest that these statements thereby affirm or are 'about' the existence of such tropes. And yet, when on a view such as GBT, statements of the form '$F\exists x\Phi x$' and '$\exists x F\Phi x$' receive a systematically different semantic treatment, then this has its basis in the thought that prefixing an existential statement by a future-tense operator is a means to cancel any ontological commitment – not only to anything that presently Φs, but also to anything that futurely Φs. There certainly are operators suited to effect such a cancellation (e.g. 'John believes that'); and until GBT's claim that future-tense operators belong to this broad kind has successfully been dislodged, insistence that truths of the form '$F\exists x\Phi x$' nonetheless require the existence of something satisfying the open sentence '$F\Phi x$' is dialectically ineffective.

Even if there was after all presently a thing of which it was true to say that it will be a newly born child a thousand years hence, its mere present existence would anyway fail to ensure that this is what can at present truly be said about it. This is why the grounding of truth is said to be in terms both of what there is and *how* what there is *is*. But now, even if Émilien presently exists, and according to GBT will continue to exist forever, why should how he presently is ground the truth of the future contingent 'Seventeen years hence, Émilien will have a ball'? That Émilien will have a ball in seventeen years' time is, plausibly, not prefigured in any of his currently exemplified properties in conjunction with his present existence – unless, that is, we allow Lucretian properties among those properties.

Consequently, its commitment to past things does not automatically put GBT in a better position to account for the grounding of truths about the past: that Nero, located in the past of now, still exists, is insufficient to ground the present truth of 'Nero was mad'. The task might become easier if one pairs GBT with an ontology of tropes or events, for arguably Nero's particularized madness is sufficient to ground the truth of 'Nero was mad'. But as long as GBT is intended to be open to those metaphysicians who reject ontological commitment to tropes or events, it is as yet no better placed to account for the truth of statements about the past than to account for the truth of statements about the future. The thought generalises to other

views: if Lucretian properties are out of bounds, and if all residents of time other than times are 3D objects that persist by enduring, even permanentists of the Williamsonian stripe face problems when asked to provide present grounds for truths about the past or future.

Suppose we continue to disallow Lucretian properties of any kind. Then there is no more mystery in the thought that the present truth of 'F(something Φs)' *will in the future be* grounded in something that then Φs, than in the thought that it is partially grounded in the present existence of something of which it is presently true to say that it will Φ, where only in the future, the truth of the latter predication will be grounded by that thing's Φ-ing. Indeed, why should it not in general be enough in order for a statement about the future to be presently true, that its present truth will in the future be grounded by something being certain ways, whose future existence and future ways of being the statement now affirms – just as it is enough in order for a statement about what happens at a distance to be true here, that at a distance, its being true here is grounded by something's happening whose distant existence and happening the statement here affirms?

Suitably generalised, this view holds that the truth of a given tensed statement at most requires that it *sometimes* be grounded in what then is something and a certain way, provided that the statement claims that there then is such a thing that then is that way (Westphal 2006; cf. also for further discussion Gallois 2004; Kierland and Monton 2007; Baia 2012; and Tallant and Ingram 2015).[3] This theoretical option shows that taking tense to be irreducible need not automatically commit one to the Lucretian answer to the truth-grounding problem, contrary to what Sider (2001: 39) concludes.

In particular then, the present truth of a statement about how, at some future time, things will be, might well be said to be, at that future time, going to be grounded by things being that way. But how, one might ask, can such future grounding *ensure the*

[3] Dummett argues that 'a proposition about what I am going to do is true in virtue of my later action' and, more generally, that '[i]t is *what is going to happen in the future* that renders our statements about the future true, when they are true. This platitude is embodied in the truth-value links' (Dummett 2004: 81, 83). Dummett concludes that a view like Broad's, according to which there is now nothing that is only ever going to happen in the future, is bound to deny that statements about the future are true (Dummett 2004: 74, 80). Contrary to what Dummett claims, however, the truth-value links merely yield that a statement about the future such as 'It will be the case that Mars is being colonized' is presently true just if it will be the case that 'Mars is being colonized' is true. It is only against the backdrop of Dummett's further contention that 'a proposition can be true only if there is something in virtue of which it is true' (Dummett 2004: 74), that we can reason from this to the conclusion that if 'It will be the case that Mars is being colonized' is presently true, there is now something in virtue of which 'Mars is being colonized' will be true, i.e. a merely future happening. The present suggestion is to replace Dummett's contention by the claim that a proposition is true at *t* only if *sometimes* there is something in virtue of which it is true at *t*. Note also that, contrary to what Dummett (2004: 80) suggests, the proponent of GBT need not treat the truth-conditions of statements about the past in any substantially different way – irrespective of the fact that, according to GBT, past things still exist (cf. also Broad 1938: 316).

present truth of such a statement? The answer to this question is pretty straightforward.

Our toy example will again be the future contingent 'One day hence, some rain will fall', which we assume to be presently true. For sake of perspicuousness, we use 'q' as shorthand for the embedded clause 'Some rain falls', '↓' for 'One day ago', and '↑' for 'One day hence'. The future contingent in question can accordingly be written as '↑q'; and the task before us correspondingly is to explain why '↑q' is presently true. Our explanation relies on two general principles, meant to be restricted to those φ which are grounding-theoretically unproblematic and naturally thought of as being in the simple present-tense:

(E1) Always, if φ is true, then ∃X(φ is true because X exist)
(E2) Always, if φ is true, then ↓('↑φ' is true) because φ is true

The plural quantifier in the first principle (E1) ranges over pluralities of the kinds of entities – things, facts, tropes, events, *etc.* – that according to one's preferred ontology, conspire to make the unproblematic statements true when they are true. Thus understood, (E1) can be assumed as common ground. So can (E2): this second principle encodes the natural enough thought that whenever a given unproblematic statement φ is true, '↑φ' was true a day ago and its having been true a day ago is grounded in the truth of φ. By the transitivity of 'because' – another plank of the common ground – (E1) and (E2) together yield:

(E3) Always, if φ is true, then ∃X(↓('↑φ' is true) because X exist)

With (E3) in place, we can now reason as follows:

(1) '↑q' is true
(2) ↑('q' is true)
(3) ↑∃X(↓('↑q' is true) because X exist)

From our initial assumption (1), we derive (2) using Tarskian conditionals; (2) and (E3) together then yield (3) which may be taken to say the same as

(4) One day hence, there will be things whose existence will explain why one day before, '↑q' was true

But reflecting on what the underlined clause in effect says, it is easy to see that (4) is tantamount to

(5) One day hence, there will be things whose existence will explain why '↑q' is true now

where 'now' takes us right back to the present time. (5) thus provides us with a successful grounding explanation of why '↑q' is presently true.

We are now also in a position to formulate a general grounding requirement that does justice to the idea that truths do not 'float free', goes beyond a simple deflationism and, at the same time, is hospitable to tensed ontologies. Thus, let ψ range over present truths of the problematic kinds – including contingent truths about the future – then we can lay down:

(GR) For all ψ, Sometimes, $\exists X$((Now, ψ is true) because X exist)

where, as before, 'Now' is a temporal operator that always shifts the time of evaluation back to the time of utterance.

(GR) allows for the possibility that a statement about the future be true now whose present truth is sometimes in the future explained by what exists then, while there exists now nothing that would explain its present truth. However it might now be suggested that this is not yet to allow for the truth of future *contingents*. Thus, it might be suggested that if a statement about the future like 'One day hence, some rain will fall' is indeed presently true, *that alone* is guarantee enough to conclude that the future is bound to be a certain way – eliminating thereby the possibility of truth for future *contingents*. In reply, it should first be noted that if we could, in principle, presently know for certain that some statement about the future is true, then given our principled inability to inspect the contingent future course of events, this would indeed suggest that that statement is not a future contingent. But such epistemic considerations are of no present concern. Rather, what is in question here is whether the present truth of a statement that will be grounded by how matters are going to stand, in any way undermines its status as a future contingent. An affirmative answer to this question, we submit, gets things backwards.

If we already built it into the very notion of a future contingent that no statement about the future can qualify as a future contingent *if its truth is already entailed by all present truths* (cf. Markosian 1995: 96), then it would trivially follow that future contingents are never true, and proponents of GBT would be ill-advised to even try to reconcile their metaphysical views with the truth of future contingents. This would still, however, allow formulation of the sensible question of whether there are any truths about the future that are not made inevitable by any present or past *things, facts, states or happenings*. Present truths about the future that will be grounded by what there will be and how it will be, may well qualify as such, as long as nothing there is or was, in conjunction with how it is or was, makes it inevitable that, in the future, there will be such grounds (Broad 1937: 204, 206; cf. also Rosenkranz 2012).[4] This is the more natural – and more neutral – conception of future contingents that leaves the question of their truth as yet open. In any case, it is this concep-

[4] Later Broad clearly distinguished between the future's being predetermined by past and present facts and its being 'predeterminate' in the sense that statements about the future have a definite truth-value (Broad 1937: 204, 206). But since he would seem to have abandoned his 1923 view by that time, he never readdressed the question of why, in the light of this distinction, GBT should be taken to be committed to denying that future contingents are ever true.

tion that we here assume when arguing for the conclusion that GBT is compatible with the truth of future contingents; and once the grounding requirement is properly construed as demanding no more than that every truth sometimes be grounded in what there then is and how it then is, such compatibility is indeed hard to deny.

Admittedly, we so far have not said enough about how the truth of the negation of a future contingent is ever grounded, and so, given that the falsity of a statement consists in the truth of its negation, we have not said enough either about how the general principle of bivalence can be reconciled with a sensibly temporalised grounding requirement on tensed truths.

We must distinguish between different cases. Consider first future contingents of the form $F_n\varphi$. Granted that time will go on for at least n units of time, the truth of $\neg F_n\varphi$ can be reduced to the truth of $F_n\neg\varphi$, whose grounding can in turn be explained along the aforementioned lines. If, by contrast, time will *not* go on for at least n units of time, then one can hold that for some m, with $0 \leq m < n$, $\neg F_n\varphi$ is true because m time-units from the present, time has come to an end, and add that m time-units from the present, it is a brute fact that time has come to an end. This grounding explanation presupposes that if time does not go on indefinitely, there is a last moment of time. This is not an entirely innocent assumption; however, note that it is compatible with time's being discrete, dense or continuous.

Consider then future contingents of the form $F\varphi$. Suppose first that time will go on for at least n units of time for any number n. Then the truth of $\neg F\varphi$ can be reduced to the truth of the general statement $\forall n F_n\neg\varphi$. To explain how the truth of generalisations may be grounded is a vexed issue; and we do not have anything new to contribute to the solution of the problem. However, once the obligation has successfully been discharged to explain how, for any given choice of n, the truth of $F_n\neg\varphi$ may sometimes be grounded, whatever difficulties remain in explaining how $\forall n F_n\neg\varphi$ might ever be grounded would seem to have nothing specifically to do with the grounding-problem posed by the truth of future contingents.[5] Suppose then that *it is not the case that* time will go on for at least n units of time for any number n. Assuming again that, if this is so, there is a last moment of time, one can hold either that $\neg F\varphi$ is true because time has already come to an end, where it is a brute fact that time has come to an end, or that, for some $0 < m$, $\neg F\varphi$ is true because (i) m time-units from the present, time has come to an end, where m time-units from the present, it is a brute fact that time has come to an end, and (ii) for all k, with $0 < k \leq m$, $F_k\neg\varphi$. The latter is a restricted generalisation, and our previous remarks about the grounding of unrestricted generalisations of type $\forall n F_n\neg\varphi$ apply *mutatis mutandis*.

The problem of accounting for the truth of generalisations, and for that of certain negative statements such as negative existentials, is a problem for everyone; and to the best of our knowledge, the principle of bivalence has never come under attack for these sorts of reasons, except perhaps from intuitionists who presuppose an

[5] The same applies to the problem of truth-grounding for certain negative statements, such as e.g. negative existentials, that we have conveniently glossed over by simply assuming that, in general, if $F\neg\varphi$ is presently true, sometimes in the future, its past truth will be grounded in whatever then grounds $\neg\varphi$.

epistemic conception of truth-grounds quite alien to the metaphysical discussion in which we are here engaged.

Modulo a solution to these latter problems, which as argued are not specific to the debate about the open future, and contrary to what Broad (1923), Dummett (2004) and others contend, GBT accordingly coheres with the principle of bivalence as applied to future contingents, which also silences the worries voiced by critics like D. C. Williams (1951b). Once the grounding requirement on truth is relaxed in these ways – see (GR) above – it will likewise apply to statements about the past: even if, according to GBT, there exist things entirely located in the past, truths about them were, in the past, grounded in how these things then were; and presentists, who deny that there exist such things, may correspondingly say that statements about the past were, in the past, grounded in what there then was and how it then was. On neither view does the grounding requirement, properly construed, force commitment to Lucretian or Cambridge-like properties, while the principle of bivalence can, to this extent at least,[6] be retained.

7.3 Indeterminism and the Open Future

As we have argued, proponents of GBT are entitled to assume classical logic, and in particular a bivalent semantics, even for future contingents. For the same reasons, there is no pressure to abandon the conception of time as being linear in favour of a conception of time as forward-branching. In the light of the appropriately relaxed grounding requirement on tensed truths, proponents of GBT can also draw a systematic distinction between being determinate in truth-value and being predetermined to be true or false: a statement like 'One day hence, some rain will fall' may not be predetermined insofar as its truth is not rendered inevitable by how things located in the present or past of now are or were in all their natural respects; but for all that, it may nonetheless be determinate in truth-value.

Proponents of GBT will therefore refuse to construe the open future in terms of any present lack of truth-value of a certain subset of statements about the future. Instead, they might construe the open future just as the phenomenon that certain statements about the future are neither predetermined to be true nor predetermined to be false – precisely those statements about the future that qualify as future contingents. Given how predetermination was being characterised above, this is also a way in which eternalists might characterise the open future. Even if always, everything always exists, and even if always, everything has its properties permanently – either because the properties in question are relations to times or because the things in question are of point-sized duration – one may nonetheless still allow that how things located in the present or past of now are, in relation to times identical to or earlier than now, does not predetermine how things located in the future of now are in relation to times later than now.

[6] Bivalence may, of course, be said to fail for other reasons, e.g. vagueness.

This may invite the following challenge: if the best that a proponent of GBT can do in order to account for the openness of the future is to claim something that can likewise be claimed by eternalists – i.e. that what will happen is not predetermined in the aforementioned sense – then they have not succeeded in capturing any interesting sense of openness. Even if we consider this too strong a claim, we take up the challenge and show that there is a more radical sense in which the future may be said to be open which is available to GBT, but not to permanentism, and hence, since eternalism implies permanentism, not to eternalism either. This sense is still perfectly compatible with the unrestricted principle of bivalence. And so, even if it should after all be true that if the future is open in any interesting sense, eternalism is incompatible with the future's being open, this thought alone does nothing to motivate the contention that future contingents are neither true nor false. Before we explain what this stronger sense is, however, let us first set out the conception of the open future that both proponents of GBT and permanentists might agree on.

The idea that the future is open is closely related to the rejection of determinism, which latter we may informally gloss as the doctrine that always how the world is, and has been, nomologically determines how the world is going to be. This thought is aptly captured by the following principle (here and below we use special variables t, t', etc. for times, which could be dispensed with, using regular variables and restricting quantification with the help of our predicate T for times):

(DET) Always, $\forall t \mathrm{G} \forall t' (t \prec t' \rightarrow$ the way the world is up to t nomologically determines the way the world is up to $t')$

Indeterminism is correspondingly understood as the negation of (DET), i.e. as the claim that

(IND) Sometimes, $\exists t \mathrm{F} \exists t' (t \prec t' \,\&$ The way the world is up to t does not nomologically determine the way the world is up to $t')$[7]

(IND) is oftentimes deemed insufficient for the open future, precisely because, as we shall see shortly, on some natural interpretations of it, (IND) is acceptable to both static and dynamic permanentists.[8] Even if this is so, however, there are other interpretations of (IND) that are not acceptable to permanentists.

[7] The formulation of (IND) allows that certain ways the world could have turned out to be were more probable than others given the way the world was up to an earlier time, and so that the laws of nature are probabilistic; and surely, any non-zero probability at least requires nomological possibility.

[8] Barnes and Cameron (2009) have even argued that a thesis like (IND) is not necessary for the open future, contrary to what we suggest here. Their reason for this claim is that it may, in some sense of 'metaphysically indeterminate', be metaphysically indeterminate what the world is like up to now, so that even fully deterministic laws will only take us from the present indeterminate world state to a later indeterminate world state: if there is any indeterminacy in the present world state, this indeterminacy simply 'may bleed over' into the subsequent world state. However, the authors' reasoning seems flawed because the sense in which the present state of the world might be indeterminate – say,

To see this, note first that both (DET) and (IND) are still pretty vague and allow for different precisifications of the phrase 'the way the world is up to *t*'. We take the latter to be a nominalisation of a statement of the form 'At *t*, ω^t' where 'ω^t' is a description of what *actually* is or used to be the case at *t*, where this description is in some pertinent sense complete. Different precisifications specify different such senses.

It should anyway be clear, though, that lest (DET) be trivialised, 'ω^t' must not contain clauses equivalent to clauses of the forms 'Fφ' or '$F_n\varphi$' (for some or all *n*), which are irreducibly future-tensed. Similarly, 'ω^t' must not contain clauses ultimately reducible to clauses of the forms '$\neg F\varphi$' or '$\neg F_n\varphi$' (for some or all *n*). To see this, consider a case in which a present configuration of particles and force fields determines that in *n* units of time, a certain event *e* with a certain determinable property Φ will occur, while no conjunction of present physical facts determines which determinate of Φ *e* will then instantiate. Suppose that in *n* units of time, *e* occurs and instantiates determinate property Φ_1. It would be cheating to try to restore coherence of this example with (DET) by conceiving 'ω^{now}' to entail '$\neg F_n\Phi_i(e)$', for each determinate of Φ, Φ_i, that is distinct from Φ_1.

Bearing these bans in mind, we can now consider one precisification of 'ω^t' according to which it yields a complete description of all the entities actually located at *t*, or at any time earlier than *t*, in terms of the natural – non-Lucretian and non-Cambridge-like – properties and relations among them that they actually instantiate at *t* or at any earlier time. To disambiguate, let 'ω^t_{loc}' be the relevant description. Since, as far as (DET) is concerned, absences may well be determining factors, we will here presume that 'ω^t_{loc}' may contain a suitable number of clauses of the form 'and that's everything located at the present time', either unembedded or embedded in the context of operators of the forms 'P', 'H' or 'At *t*″', with $t' \prec t$. The corresponding determinist claim then is

$(\mathrm{DET}_{\mathrm{loc}})$ Always, $\forall t \mathrm{G} \forall t' (t \prec t' \rightarrow \Box(\mathrm{At}\ t, \omega^t_{loc} \rightarrow \mathrm{At}\ t', \omega^{t'}_{loc}))$

where '$\Box$' encodes nomological necessity. Both static and dynamic permanentism and GBT are compatible with the truth of $(\mathrm{DET}_{\mathrm{loc}})$.[9] Let us therefore now turn to the question of whether either of these two views can be made to cohere with the corresponding form of indeterminism, i.e.

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the sense in which it is indeterminate which, if any, cell has survived fission – is not the sense in which the future is said to be indeterminate by being open: by the authors' own lights, the future is open in a sense in which the present and past are not (see Rosenkranz 2013: 69, for discussion).

[9] McTaggart (1927: §337) rightly observed that this would commit proponents of GBT to the truth of statements about the future – something at odds with what Broad takes himself to be committed to (Broad 1923: 73). As we have argued above, however, Broad (1923) is mistaken when he contends that proponents of GBT must treat statements about the future as being neither true nor false: even future contingents can be regarded as bivalent, and some as true, quite consistently with GBT (cf. also Broad 1937: 206).

(IND_{loc}) Sometimes, $\exists t \mathrm{F} \exists t' (t \prec t' \,\&\, \Diamond(\mathrm{At}\ t, \omega^{t}_{loc} \,\&\, \neg\, \mathrm{At}\ t', \omega^{t'}_{loc}))$

For simplicity's sake, assume that *now* $\prec$ *t*, and that while actually both 'Now, ω^{now}_{loc}' and 'At *t*, ω^{t}_{loc}' hold, '$\Diamond$(Now, ω^{now}_{loc} & ¬At *t*, ω^{t}_{loc})' also holds. In application to this particular case, we can distinguish at least three grades of (IND_{loc}) depending on the counterfactual scenarios that can consistently be claimed to bear witness to the possibility claim, *viz.*

(I) Now, ω^{now}_{loc}, but at *t*, some of the things actually located at *t* are located at *t* and have properties distinct from the ones they actually have at *t*

(II) Now, ω^{now}_{loc}, but while *t*, and times later than *t*, sometimes exists, always, for all times *t′*, such that either *t* = *t′* or *t* $\prec$ *t′*, no entity distinct from *t′* is ever located at *t′*

(III) Now, ω^{now}_{loc}, but neither *t* nor any time later than *t* ever exists

It would seem that both static and dynamic permanentism and GBT can allow for all three grades of (IND_{loc}). Note, however, that on any eternalist ontology exclusively of spatiotemporally individuated events that also have all their non-spatiotemporal properties essentially, (I) is after all no option. Such an ontology would still allow for (III) to hold, though, and also for (II) to hold provided that such an ontology likewise includes instants of time.

According to another precisification of 'ω^{t}', it yields a complete description of all the entities actually *existing* at *t*, or at any time earlier than *t*, in terms of the natural – non-Lucretian and non-Cambridge-like – properties and relations among them that they actually instantiate at *t* or at any earlier time. To disambiguate, let 'ω^{t}_{ex}' be the relevant description. Since, as far as (DET) is concerned, absences may well be determining factors, we will here presume that 'ω^{t}_{ex}' may contain a suitable number of clauses of the form 'and that's everything existing at the present time', either unembedded or embedded in the context of operators of the forms 'P', 'H' or 'At *t″*', with *t′* $\prec$ *t*.

On this precisification, the corresponding determinist and indeterminist claims accordingly are:

(DET_{ex}) Always, $\forall t \mathrm{G} \forall t' (t \prec t' \rightarrow \Box(\mathrm{At}\ t, \omega^{t}_{ex} \rightarrow \mathrm{At}\ t', \omega^{t'}_{ex}))$

(IND_{ex}) Sometimes, $\exists t \mathrm{F} \exists t' (t \prec t' \,\&\, \Diamond(\mathrm{At}\ t, \omega^{t}_{ex} \,\&\, \neg\, \mathrm{At}\ t', \omega^{t'}_{ex}))$

These readings of determinism and indeterminism may admittedly be less familiar; however, they still fit the moulds of (IND) and (DET) and involve admissible precisifications of the phrase 'the way the world is up to *t*'.

Assume that *now* $\prec$ *t*, and that while actually both 'Now, ω^{now}_{ex}' and 'At *t*, ω^{t}_{ex}' hold, '$\Diamond$(Now, ω^{now}_{ex} & ¬At *t*, ω^{t}_{ex})' also holds. Again, in application to this particular case, we can distinguish three grades of (IND_{ex}) depending on whether the counter-

factual scenarios described by (I′) to (III′) below can consistently be claimed to bear witness to the possibility claim '$\Diamond$(Now, $\omega^{now}{}_{ex}$ & ¬At t, $\omega^{t}{}_{ex}$)':

(I′) Now, $\omega^{now}{}_{ex}$, but at t, some of the things actually located at t are located at t and have properties distinct from the ones they actually have at t

(II′) Now, $\omega^{now}{}_{ex}$, but while t, and times later than t, sometimes exists, always, for all times t', such that either $t = t'$ or $t \prec t'$, no entity distinct from t' is ever located at t'

(III′) Now, $\omega^{now}{}_{ex}$, but neither t nor any time later than t ever exists

As in the case of (I), on any eternalist ontology exclusively of spatiotemporally individuated events that also have all their non-spatiotemporal properties essentially, (I′) is unavailable. For permanentists, and hence eternalists, (II′) is no option, unless they presume that all things other than times that are actually located at t, or at a time later than t, can all be located at times earlier than t – which would seriously constrain their ontology. Since according to permanentists, now, t and all times later than t exist, (III′) is no option for them, irrespective of what further assumptions they might be willing to make about the existents that they are, *qua* permanentists, committed to.

By contrast, GBT is unproblematically consistent with (I′) to (III′). So there is at least one interpretation of (IND) – (IND_{ex}) of grade (III′) – which coheres with GBT but is clearly not open to permanentists, and another – (IND_{ex}) of grade (II′) – which coheres with GBT but is very unlikely to be compatible with permanentism.

Accordingly, if the task was to come up with a conception of the open future available on GBT but unavailable on any permanentist views, the conception that allows for the possibility of either type of scenario will fit the bill. And yet, this conception of the open future is perfectly compatible with the bivalence of all future-tensed statements, including future contingents.

Presentists, too, can avail themselves of these conceptions of the openness of the future, without being forced to reject, or systematically restrict, bivalence. If they do so, however, they will be forced to say that the past is open in the very same sense, in which case they can no longer account for the asymmetry between the future as open and the past as fixed which underlies the intuitions that fuel attempts to come up with such conceptions in the first place (see, however, Markosian 1995). GBT, of course, faces no such problem, because on GBT, what existed in the past still exists. This makes GBT much better positioned to invoke (IND_{ex}) of grade (II′) or (III′) in the attempt to give an account of the open future.